

```
!pip install faiss-cpu
```

```
Requirement already satisfied: faiss-cpu in  
/usr/local/lib/python3.12/dist-packages (1.13.2)  
Requirement already satisfied: numpy<3.0,>=1.25.0 in  
/usr/local/lib/python3.12/dist-packages (from faiss-cpu) (2.0.2)  
Requirement already satisfied: packaging in  
/usr/local/lib/python3.12/dist-packages (from faiss-cpu) (26.0)
```

```
import pandas as pd  
import numpy as np  
import faiss  
from sklearn.feature_extraction.text import TfidfVectorizer  
  
products = pd.read_csv(r"/content/drive/MyDrive/amazon_products.csv")  
categories =  
pd.read_csv(r"/content/drive/MyDrive/amazon_categories.csv")  
  
print(products.head())  
print(categories.head())
```

	asin	title \
0	B014TMV5YE	Sion Softside Expandable Roller Luggage, Black...
1	B07GDLQXV	Luggage Sets Expandable PC+ABS Durable Suitcas...
2	B07XSCCZYG	Platinum Elite Softside Expandable Checked Lug...
3	B08MVFKGJM	Freeform Hardside Expandable with Double Spinn...
4	B01DJLKZBA	Winfield 2 Hardside Expandable Luggage with Sp...

	imgUrl \
0	https://m.media-amazon.com/images/I/815dLQKYIY...
1	https://m.media-amazon.com/images/I/81bQlm7vf6...
2	https://m.media-amazon.com/images/I/71EA35zvJB...
3	https://m.media-amazon.com/images/I/91k6NYLQyI...
4	https://m.media-amazon.com/images/I/61NJoaZcP9...

	productURL	stars	reviews	price
listPrice \				
0	https://www.amazon.com/dp/B014TMV5YE	4.5	0	139.99 0.00
1	https://www.amazon.com/dp/B07GDLQXV	4.5	0	169.99 209.99
2	https://www.amazon.com/dp/B07XSCCZYG	4.6	0	365.49 429.99
3	https://www.amazon.com/dp/B08MVFKGJM	4.6	0	291.59 354.37
4	https://www.amazon.com/dp/B01DJLKZBA	4.5	0	174.99 309.99

	category_id	isBestSeller	boughtInLastMonth
0	104	False	2000
1	104	False	1000

2	104	False	300
3	104	False	400
4	104	False	400

	id	category_name
0	1	Beading & Jewelry Making
1	2	Fabric Decorating
2	3	Knitting & Crochet Supplies
3	4	Printmaking Supplies
4	5	Scrapbooking & Stamping Supplies

```
print(products.columns)
```

```
Index(['asin', 'title', 'imgUrl', 'productURL', 'stars', 'reviews',
      'price',
      'listPrice', 'category_id', 'isBestSeller',
      'boughtInLastMonth'],
      dtype='object')
```

```
print(categories.columns)
```

```
Index(['id', 'category_name'], dtype='object')
```

```
data = pd.merge(products, categories, left_on="category_id",
right_on="id")
```

```
print(data.head())
```

	asin	title \
0	B077CBD2RX	For Honda Odyssey Rear Reflector Assembly Unit...
1	B0BGQQLGDM	Aveeno Repairing CICA Foot Mask with Prebiotic...
2	B0B7NP4RLN	PATIKIL 3.4 Inch Dia Key Organizer Keychain, 1...
3	B0BDYQP3KL	Npt Pipe Die Hex Die Carbon Steel Rethreading ...
4	B0C3GVWNX3	Extra Large Travel Backpack, 45L Large Laptop ...

	imgUrl \
0	https://m.media-amazon.com/images/I/61tzY2po-N...
1	https://m.media-amazon.com/images/I/71K7mk5TFg...
2	https://m.media-amazon.com/images/I/71ZhZ704z6...
3	https://m.media-amazon.com/images/I/61g3P-0CEX...
4	https://m.media-amazon.com/images/I/91udk0p281...

	productURL	stars	reviews	price
listPrice \				
0	https://www.amazon.com/dp/B077CBD2RX	4.7	0	25.14
0.00				
1	https://www.amazon.com/dp/B0BGQQLGDM	4.7	262	20.34
22.99				
2	https://www.amazon.com/dp/B0B7NP4RLN	0.0	0	7.49
7.99				
3	https://www.amazon.com/dp/B0BDYQP3KL	4.4	0	21.99

```
0.00
4 https://www.amazon.com/dp/B0C3GVWNX3 4.9 0 32.99
49.99
```

	category_id	isBestSeller	boughtInLastMonth	id	\
0	15	False	0	15	
1	51	False	1000	51	
2	138	False	0	138	
3	153	False	0	153	
4	65	False	50	65	

	category_name
0	Heavy Duty & Commercial Vehicle Equipment
1	Foot, Hand & Nail Care Products
2	Commercial Door Products
3	Cutting Tools
4	Laptop Accessories

```
data["text"] = data["title"].astype(str) + " " +
data["category_name"].astype(str)
```

```
print(data["text"].head())
```

```
0 For Honda Odyssey Rear Reflector Assembly Unit...
1 Aveeno Repairing CICA Foot Mask with Prebiotic...
2 PATIKIL 3.4 Inch Dia Key Organizer Keychain, 1...
3 Npt Pipe Die Hex Die Carbon Steel Rethreading ...
4 Extra Large Travel Backpack, 45L Large Laptop ...
Name: text, dtype: object
```

```
vectorizer = TfidfVectorizer(stop_words="english")
vectors = vectorizer.fit_transform(data["text"])
product_vectors = vectors.toarray().astype("float32")
```

```
print(product_vectors.shape)
```

```
(5000, 14612)
```

```
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```

```
import faiss
```

```
dimension = product_vectors.shape[1]
```

```
index = faiss.IndexFlatL2(dimension)
```

```
index.add(product_vectors)
```

```
print("Total products indexed:", index.ntotal)
```

```
Requirement already satisfied: faiss-cpu in
```

```
/usr/local/lib/python3.12/dist-packages (1.13.2)
```

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Requirement already satisfied: numpy<3.0,>=1.25.0 in
```

```
/usr/local/lib/python3.12/dist-packages (from faiss-cpu) (2.0.2)
```

```
Requirement already satisfied: packaging in
```

```
/usr/local/lib/python3.12/dist-packages (from faiss-cpu) (26.0)
Total products indexed: 5000
```

```
def recommend(product_name, k=5):

    matches = data[data["title"].str.contains(product_name,
case=False)]

    if len(matches) == 0:
        print("Product not found")
        return

    idx = matches.index[0]

    query_vector = product_vectors[idx].reshape(1, -1)

    distances, indices = index.search(query_vector, k+1)

    print("\nProducts similar to:", data.iloc[idx]["title"])
    print("-----")

    for i in indices[0][1:]:
        print(data.iloc[i]["title"], "-", data.iloc[i]
["category_name"])

recommend("headphones")
```

```
Products similar to: UX530 Wireless Earbuds for Motorola Moto G Stylus
5G with Immersive Sound True 5.0 Bluetooth in-Ear Headphones with
2000mAh Charging Case - Stereo Calls Touch Control IPX7 Sweatproof
Deep Bass
```

```
-----
Clear Sound, True Wireless, Ultra Portable, Water Proof, Touch Control
Earbuds. - Headphones & Earbuds
```

```
Wireless Earbuds, Bluetooth 5.3 Headphones Noise Cancelling with
Charging Case, IPX7 Waterproof Stereo Earphones in-Ear Earbud with
Microphone for Android Cell Phone Gaming Computer Laptop Sport -
Headphones & Earbuds
```

```
RHA Trueconnect - Carbon Black: True Wireless Earbuds with Bluetooth 5
& Sweatproof for Sport Activity - Headphones & Earbuds
```

```
MURPISO Wireless Earbuds, Active Noise Cancelling Bluetooth5.2 Deep
Bass Headphones with 4 Mics, 35 Hours Playtime, IPX6 Waterproof
Earphones, in-Ear Bluetooth Earbuds for iPhone Android Black -
Headphones & Earbuds
```

```
Wireless Earbuds, Bluetooth 5.3 Earbuds in-Ear, 50Hrs Playtime True
Wireless Earphones Build in ENC and ANC Noise Cancelling Mic, with
Charging Case, IP7 Waterproof Sport EarBuds for iPhone & Android -
Headphones & Earbuds
```

```
recommend("samsung galaxy")
```

Products similar to: Samsung Galaxy Tab A7 Lite Case 8.7" w/Glass
Screen Protector Stand&Hand Strap | Heavy Duty Rugged Silicone Kids
Case for Galaxy Tab A7 Lite 2021 SM-T220/T225/T227U | Black

Case Compatible with Galaxy Tab S8 11 Inch 2022 (SM-X700/SM-X706)/S7
2020 (SM-T870/T875) with S-Pen Stand, Slim & Lightweight Magnetic Case
with Auto Sleep/Wake for Galaxy Tab S8, Black - Tablet Accessories
Targus FieldReady Case for Samsung Galaxy Tab Active3 - Black - Tablet
Accessories

USB Charging Flex Cable Board Port Charge Connector Dock Socket Jack
Plug Replacement for Samsung Galaxy Tab A7 10.4 inch T500 T505 -
Tablet Replacement Parts

Screen Replacement for Samsung Galaxy SM-T280 LCD Display Touch
Digitizer Panel Full Assembly Compatible for Tab A 7.0 WiFi Tablet
(Not for 3G Version & T285 & No Earpiece Hole) (Black) - Tablet
Replacement Parts

Supershieldz (3 Pack) Designed for Samsung Galaxy Tab A8 10.5 inch
Screen Protector, High Definition Clear Shield (PET) - Computers